

VendorMind

Autonomous WhatsApp Conversational Commerce

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URL: vendormind-z.web.app

Executive Summary

VendorMind is a next-generation conversational commerce platform that enables merchants to sell products and accept secure payments directly inside WhatsApp. By combining a brand-tailored LLM sales assistant with **Nomba Online Checkout**, VendorMind removes checkout friction entirely—eliminating app downloads, registrations, and external web redirects for consumers.

The Problem

Mobile commerce is plagued by several points of friction:

- **App Fatigue:** Over 80% of consumers abandon purchases when forced to download a new application or register a web account.
- **Manual Overhead:** Small-to-medium businesses spend hours manually updating inventory, checking stock levels, calculating shipping rates, and verifying payment alerts in chat.
- **Disjointed Payment Experiences:** Direct payments in messaging apps are non-existent, forcing consumers to navigate through slow web interfaces.

The Solution: VendorMind

VendorMind connects directly to a vendor's WhatsApp number and handles transactions autonomously:

1. **Semantic Catalog Browsing:** Customers query the catalog using natural, casual sentences (e.g. *"Do you have anything sweet?"*), processed via vector embeddings.
2. **Real-time Cart Management:** Automated addition/removal of products and temporary stock reservations.
3. **Direct Checkout Links:** Instantly creates secure Nomba Checkout Order URLs containing the exact cart total in Naira.
4. **Payment Webhooks:** Automatically listens for Nomba's `payment_success` webhooks to mark orders paid, release reserves, update inventory levels, and send receipts.

Core System Features

1. Hybrid Merchant Dashboard

A premium PWA dashboard with a dark/light switcher that features a vertical onboarding wizard, WhatsApp pairing/QR codes, Excel bulk catalog uploading, and real-time wallet tracking.

2. Interactive Human Handoff

If a customer requires human assistance or if the vendor intervenes, the bot automatically pauses and locks itself out, transitioning the session to a human operator in the live chat inbox.

3. Intelligent Conversation Guardrails

A dedicated classification model separates business inquiries from casual personal chat. If a family member or friend messages the vendor, the bot stands down immediately without charging any processing credits.

Tech Stack

- **Frontend:** React, Vite, CSS variables, PWA (Vite PWA plugin), Lucide Icons, Lenis Smooth Scroll
- **Backend REST API:** Fastify (Node.js/TypeScript)
- **Queue & Event Processing:** BullMQ & Redis (5 concurrent background workers)
- **Database & ORM:** PostgreSQL with Prisma ORM
- **WhatsApp integration:** Baileys Multi-Device Socket SDK
- **AI Orchestration:** Anthropic Claude-Sonnet-4.6, Google Gemini 2.5, and Groq Audio Whisper
- **Payments API:** Nomba Checkout & Auth Gateway